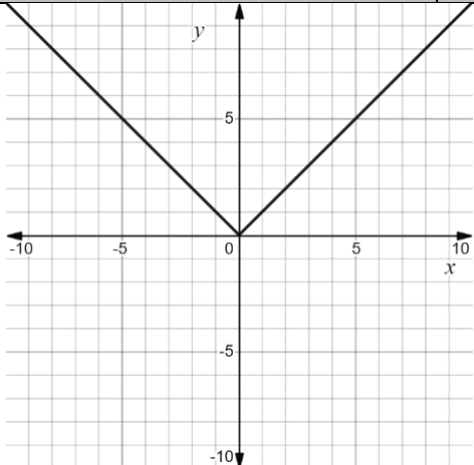


1. Select the function family for each.

	Function	Function Types
a.		☒ Absolute Value Function ☐ Cubic Function ☐ Linear Function ☐ Square Root Function ☐ Cube Root Function ☐ Exponential Function ☐ Quadratic Function
b.	$y = x^2$	☐ Absolute Value Function ☐ Cubic Function ☐ Linear Function ☐ Square Root Function ☐ Cube Root Function ☐ Exponential Function ☒ Quadratic Function

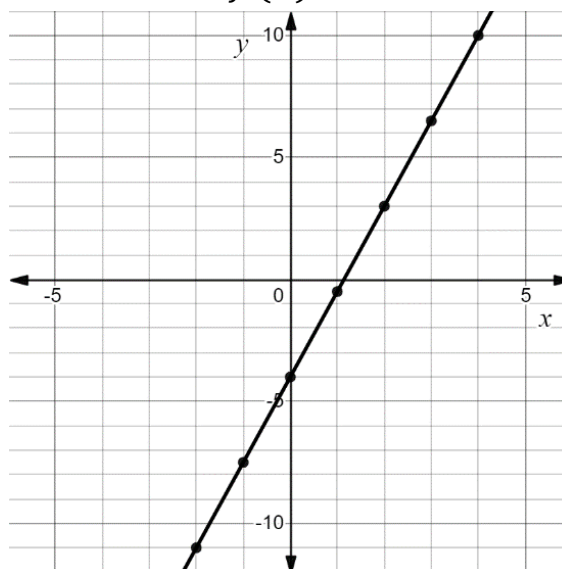
2. Find each value using the graph of the function $f(x)$.

a. $f(-2) = -11$

b. $f(4) = 10$

c. Find x when $f(x) = -7.5$.

$x = -1$



3. Use the table of values representing $g(x)$ to find each value.

a. $g(-6) = 1$

b. $g(4) = -6$

c. Find x when $g(x) = 0$.

$x = -5$

x	$g(x)$
-8	4
-6	1
-5	0
-3	-2
0	-4
2	-5
4	-6

4. The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, that U.S. passengers traveled on a train for the period 1970 to 2000, where x is the number of years since 1970.

- a. Find $M(10)$. Show your work for partial credit.

$M(10) =$	5079
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- b. Interpret $M(10)$ for the given context.

Answers may vary.

In 1980, U.S. passengers traveled 5079 millions of miles on a train.

5. The function $P(w) = 5,673w - 1,354$ models the relationship between a diamond's weight, w , in carats, and its price, P , in dollars.

Find $P(2)$. Show your work for partial credit.

$P(2) =$	9992
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6. Find the value of the variable that makes each equation true. For partial credit, show your work or explain your thinking.

Equation	Value of the Variable
a. $3^{\frac{21}{8}} \cdot 3^{\frac{11}{8}} = 3^x$	$x = 4$
b. $2^{24} = (2^y)^3$	$y = 8$
c. $61^{\frac{1}{3}} = \sqrt[k]{61}$	$k = 3$
d. $\frac{4^{\frac{3}{5}}}{4^{\frac{18}{5}}} = 4^n$	$n = -3$

7. Six expressions are shown.

A	B	C	D	E	F
$\sqrt{9^3}$	$(9^3)^{\frac{1}{2}}$	$\left(9^{\frac{1}{3}}\right)^2$	$(9)^{\frac{2}{3}}$	$(9)^{\frac{3}{2}}$	$(\sqrt{9})^3$

Select all of the expressions that are equivalent to 27.

☐ Expression A

☐ Expression B

☐ Expression C

☐ Expression D

☐ Expression E

☐ Expression F

8. Six expressions are shown.

A	B	C
$8\sqrt{2}$	$4\sqrt{8}$	$2 \cdot (16^{\frac{1}{2}})^3$
D	E	F
$(16)^{\frac{3}{2}} \cdot (2)^{\frac{1}{2}}$	$(4)^{\frac{3}{2}} \cdot (2)^{\frac{1}{2}}$	$2\sqrt{8}$

Select all of the expressions that are equivalent to $\sqrt{128}$.

☐ Expression A

☐ Expression B

☐ Expression C

☐ Expression D

☐ Expression E

☐ Expression F

9. Evaluate each expression. Show your work for partial credit.

a. $(36)^{\frac{1}{2}} \cdot (3^3)$	b. $(6^{-1}) \left(9^{\frac{3}{2}}\right)$				
<table border="1" style="width: 100%;"><tr><td style="background-color: #cccccc; width: 20%; text-align: center; padding: 5px;">Value</td><td style="text-align: center; padding: 5px; color: red;">162</td></tr></table>	Value	162	<table border="1" style="width: 100%;"><tr><td style="background-color: #cccccc; width: 20%; text-align: center; padding: 5px;">Value</td><td style="text-align: center; padding: 5px; color: red;">$\frac{9}{2}$</td></tr></table>	Value	$\frac{9}{2}$
Value	162				
Value	$\frac{9}{2}$				

10. Write a different equivalent expression for the expression.

$$\sqrt{75} + \sqrt{27} - \sqrt{48} = 4\sqrt{3}$$

11. Find the value of each expression. Show your work for partial credit.

a. $\sqrt{12} \cdot \sqrt{12}$	b. $3\sqrt{72} + 7\sqrt{8}$				
<table border="1" style="width: 100%;"><tr><td style="background-color: #cccccc; width: 20%; text-align: center; padding: 5px;">Value</td><td style="text-align: center; padding: 5px; color: red;">12</td></tr></table>	Value	12	<table border="1" style="width: 100%;"><tr><td style="background-color: #cccccc; width: 20%; text-align: center; padding: 5px;">Value</td><td style="text-align: center; padding: 5px; color: red;">$32\sqrt{2}$</td></tr></table>	Value	$32\sqrt{2}$
Value	12				
Value	$32\sqrt{2}$				